

and preventive maintenance check-ups). If maintained well, their parts don't fail or wear out for almost a decade, and thus robots' operational costs mainly include power costs and grease cost.

2. Under-utilisation of human resources. Robots can be used to executive repetitive tasks like loading and unloading of expensive CNC machines, freeing the human operators to take up value-addition jobs.

3. Excessive overheads. The actual labour cost is typically 1.5 to two times of what is paid to an employee. This can be attributed to the cost of supervisors and managers who are required to manage the labour, human resources staff, and other resources like break rooms, restrooms and cafeteria, in addition to expenses related to gratuity, bonus, insurance and perks provided to employees.

"While investing in robotics, the first thing that comes to mind is the replacement of labour. However, in reality, labour costs are just a small part of the entire consideration," explains Arjun.

Advantages of robotics in manufacturing

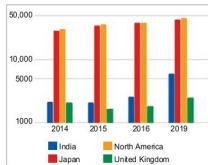
According to The International Federation of Robotics, "Automotive sector is the most significant consumer of industrial robots, buying 40 per cent of the total global output. Since the beginning of this decade, sales to carmakers have increased at an average pace of 20 per cent a year."

Examples below show how manufacturing productivity increases after deploying robotics:

1. An automotive parts manufacturer manufactured 9000 parts per day using a forging press. The cost of the forging press was more than ₹ 10 million. When the manufacturer installed a robot to feed the press, the production immediately increased from 9000 to more than 12,000 parts per day. The cost of the

robotic solution installed was about ₹ 2 million, whereas immediate direct savings from increased machine utilisation alone were equivalent to ₹ 3 million, considering 30 per cent performance boost of the ₹ 10-million press. Additionally, the manpower requirement reduced to less than one-third.

2. A pumpset manufacturer in Haryana wanted to increase the productivity of their CNC machines. At their manufacturing plant, parts



Estimated units of industrial robots in selected countries (Source: <https://ifr.org>)

were manufactured using three CNC machines, each costing ₹ 1.5 million. The manufacturer chose to replace nine workers with a single robot (costing ₹ 2.5 million) to feed all the three CNC machines, which increased productivity by 25 per cent. This increase in productivity alone translated to ₹ 1.125 million savings as the total cost of three CNC machines was ₹ 4.5 million. Also, the number of people required for three shifts reduced from nine to three, which added to cost savings.

Return on investment in both the afore-mentioned cases was realised within two years.

Challenges in implementation

"India is a slow adopter of robotics, primarily because indigenous robotics manufacturers don't get basic parts like motors and certain MSME voice recognition sensors in the Indian market and they have to import these from Japan. This increases the cost of a basic robot from ₹ 300,000

to ₹ 600,000," says Purushothaman.

Also, despite the increasing utility of robots with AI integration, these are still very complex machines.

"Highly skilled manpower is required to implement and maintain robots. This 'skilled manpower' is mostly provided by the robotics company, which increases the investment cost even further," cites Arjun.

"New-generation robots, called 'cobots,' come with an intuitive user interface and additional features, which allows any person to control them through a modern smartphone with very little training. But these cobots are more expensive and have just started entering the market. In India, their implementation is further limited due to lack of R&D," he adds.

What does the future hold?

According to Ajay Gadora, director, Ennovate Labs, "In India, use of robotics in the industrial sector is fast catching up, especially for tasks such as machine-level contingency monitoring and sharing the industrial workload without posing a threat to already existing manpower. Even the healthcare sector in India is rapidly adopting automation/robotics."

According to a research by BCG Group in the US, employing a human welder in a factory costs about \$25 per hour, which drops to just \$8 per hour for a robot, including installation, operating and maintenance costs. By 2030, the operating cost per hour for a robot doing similar welding tasks could plunge to as little as \$2 when improvements in performance are factored in.

So, "instead of viewing robotics as a technology that cuts jobs, we should rather embrace it as an opportunity helping us to focus on more strategic and creative tasks," sums up Amit Maheshwari. **EFY**

—Nidhi Arora, executive editor, EFY